Multifactor Models and the Arbitrage Pricing Theory

Jinfan Zhang

Chinese University of Hong Kong (Shenzhen)

Overview

- Systematic and Non-Systematic Risks.
  - Single factor (market) model
  - Multifactor models
- Tracking Portfolios
- Arbitrage and Arbitrage Pricing

Systematic versus Non-systematic Risk

inflation↑ $\frac{D_{t+1}\uparrow}{(r_f + r_M \text{ premium}) - g}$ (r_f↑)

- Asset returns can be decomposed into two sources:
  1. A small number of common factors which proxy for economic events that affect a large number of assets.
     - Example: changes in interest rates, inflation, and productivity.
     - These represent **Systematic risk**, which cannot be diversified away.
  2. A risk component that is unique to the asset.
     - Example – new product innovations, changes in management, lawsuits, labor strikes, etc.
     - These are **Non-systematic idiosyncratic, or firm-specific risk**, which typically is diversifiable.

Factor models

- **Factor models** are equations, that break down an asset's return into systematic component and idiosyncratic component

- One Factor Model
  - Only one systematic factor (e.g., market factor)

- Multi-Factor Model
  - Mutiple systematic factors

The One Factor Model

- Sometimes convenient to think of the one factor as the market, hence, we can refer to this model as the **market model**.

- However, the market model and the CAPM are NOT the same.

Market Model

- To illustrate this model, let's regress the returns of a stock, say Microsoft, on the market index.

- The regression equation is the following:

$$\tilde{r}_{Micro} = \alpha_{Micro} + \beta_{Micro}\tilde{R}_{S\&P} + \tilde{\epsilon}_{Micro}$$

$$\text{dependent var.} = \text{intercept} + \left(\text{slope}\right)\text{independent var.} + \text{residual}$$

- By the properties of regression, the residuals have a mean of zero, and are uncorrelated with the returns on the S&P 500.

Market Model

- Thus, the return on Microsoft is decomposed into:
  1. A constant term.
  2. Movements in the S&P 500 index return.
  3. Movements in a component unrelated to market movements.

- Decomposing the variance of Microsoft

$$\begin{aligned} var(\tilde{r}_{Micro}) &= var(\beta_{Micro}\tilde{R}_{S\&P}) + var(\tilde{\epsilon}_{Micro}) \\ &\quad +2cov(\beta_{Micro}\tilde{R}_{S\&P}, \tilde{\epsilon}_{Micro}) \\ &= var(\beta_{Micro}\tilde{R}_{S\&P}) + var(\tilde{\epsilon}_{Micro}) + 0 \end{aligned}$$

Market Model

- Variance decomposition

$$var(\tilde{r}_{Micro}) = \beta^2_{Micro} var(\tilde{R}_{S\&P}) + var(\tilde{\epsilon}_{Micro})$$

$$= \text{systematic risk} + \text{non-systematic risk}$$
$$= \text{non-diversifiable risk} + \text{diversifiable risk}$$
$$= \text{market risk} + \text{unique or firm-specific risk}$$

Market Model vs. CAPM

- The difference between the Market Model and the CAPM:
  - The CAPM describes **expected** or **mean returns**.
  - The market model describes **realized returns.** (at every moment)

- If the market model is true, then

$$\tilde{r}_i = \alpha_i + \beta_i \tilde{R}_M + \tilde{\epsilon}_i \qquad \forall i$$

- Key assumption: The residuals, $\tilde{\epsilon}_i$, are independent across stocks. Hence, they are firm specific, and diversifiable

Multi-factor Model

- However, aren't there other sources of risk that affect a group of assets, which are not picked up by an asset's sensitivity to the market?

- Example, interest rate changes.
  - Changes in interest rates affect the return on Microsoft, as well as almost all assets in the economy.
  - Some of these effects will be accounted for by the market factor, but not all.
  - The remaining effects show up in the residual of Microsoft.
  - But, because interest rate changes are not "firm specific"

Multi-factor Model

∵ Not independent now (都与 interest rate 相关)

- Hence, residuals cannot be diversified away. We can no longer view them as diversifiable risk
- How can we account for this common factor?
- **Multifactor Models**: Add additional common factors to the return process:

after given $B_1 \cdots B_K$, under No arbitrage, α is fixed.

$$\tilde{r}_i = \alpha_i + \beta_{i1}\tilde{F}_1 + \beta_{i2}\tilde{F}_2 + \ldots + \beta_{iK}\tilde{F}_K + \tilde{\epsilon}_i$$

$$\tilde{r}_i = \alpha_i + \sum_{k=1}^{K} \beta_{ik}\tilde{F}_k + \tilde{\epsilon}_i$$

$E(\tilde{r}_i) = \alpha_i$

zero mean.

  - $\tilde{F}$'s are a small number of **common factors** (1...K), for which different assets have different sensitivities, or loadings, β_{ik}

Multi-factor Model (cont'd)

- With enough common factors, the residuals are now truly firm-specific. (independent).
- We can think of the common factors as representing new information about macroeconomic variables.
  - Example: interest rates, industrial production, inflation, market volatility.
- Think of these as **surprises** in the macroeconomy (i.e., they cannot be predicted).
- Therefore, the means of the factors must be **zero**.
- This means, that the α's are the **expected returns** of the asset. Different from CAPM $\Rightarrow$ α expected to be zero

Tracking Portfolios

- A **tracking portfolio** is a portfolio that is designed to "track" the movements of an asset or investment.
- Under a factor model, this portfolio is designed to **load** on all the factors in the same way as the asset.
- That is , the tracking portfolio and the asset have the same β_{ik}

Tracking Portfolios

- Applications
  1. Hedging – to track exposures to interest rates, currencies, etc.
  2. Benchmarking – to minimize deviations from a benchmark such as the S&P 500 index.
  3. Arbitrage – to exploit mispricing among securities or investments.

Designing tracking portfolios

1. Determine the number of factors and their identity (very difficult!)

2. Compute factor betas or loadings

3. Form a portfolio which has the same factor betas as the investment you wish to track.
   - This can be done by setting up one equation for each factor beta, and then solving the equations simultaneously.

One example

- Imagine a two factor model.
- Goal: Track an asset, which has a loading of 2 on the first factor and a loading of 1 on the second factor
- Using the following three securities.

$$\begin{aligned}\tilde{r}_A &= 0.04 + 2\tilde{F}_1 - 2\tilde{F}_2 + \tilde{\epsilon}_A \\ \tilde{r}_B &= 0.03 - 1\tilde{F}_1 + 2\tilde{F}_2 + \tilde{\epsilon}_B \\ \tilde{r}_C &= 0.11 + 3\tilde{F}_1 + \tilde{\epsilon}_C\end{aligned}$$

One example (cont'd)

- We need to find a portfolio with weights that sum to one such that the portfolio weighted average of the betas equals 2 for factor one and 1 for factor two.

$$2\,w_A - 1\,w_B + 3\,w_C = 2$$
$$-2\,w_A + 2\,w_B = 1$$
$$w_A + w_B + w_C = 1$$

→ $w_A = -0.2;\ w_B = 0.3;\ w_C = 0.9$

- Note: we need K+1 securities to track an investment under a K factor model.

Pure factor portfolios

- Tracking portfolios designed to track a factor.
- Portfolio has a loading of 1 on the factor, and loadings of zero on all other factors.
- Note: these are well-diversified portfolios. Hence, they have no firm-specific risk.

Factor 1 portfolios

- Portfolio construction

$$2\,w_A - 1\,w_B + 3\,w_C = 1$$
$$-2\,w_A + 2\,w_B = 0$$
$$w_A + w_B + w_C = 1$$

→ $w_A = 0.4;\ w_B = 0.4;\ w_C = 0.2$

- Expected return
  - 0.4*0.04 + 0.4*0.03 + 0.2*0.11=5%

Factor 2 portfolios

- Portfolio construction

$$2\,w_A - 1\,w_B + 3\,w_C = 0$$
$$-2\,w_A + 2\,w_B = 1$$
$$w_A + w_B + w_C = 1$$

→ $w_A = 0.2;\ w_B = 0.7;\ w_C = 0.1$

- Expected return
  - 0.2*0.04 + 0.7*0.03 + 0.1*0.11=4%

Risk free asset

- Portfolio construction

$$2\,w_A - 1\,w_B + 3\,w_C = 0$$
$$-2\,w_A + 2\,w_B = 0$$
$$w_A + w_B + w_C = 1$$

→ $w_A = 0.6;\ w_B = 0.6;\ w_C = -0.2$

- Asset return
  - 0.6*0.04 + 0.6*0.03 - 0.2*0.11=2%

Designing pure factor portfolios

- Factor 1:
  - Expected return is 5%,
  - Premium is 3%

- Factor 2:
  - Expected return is 4%,
  - Premium is 2%

Interpretation

- The risk premiums of factor portfolios are usually denoted as λ_k for factor k.

- Premium depends on aggregate supply of the factor in the economy and the tastes of investors.

Interpretation (cont'd)

- We can use the factor portfolios to track any investment.
- To track the asset in the earlier example, we would simply buy two of factor portfolio one and buy one of factor portfolio two.
- Since pure factor portfolios are themselves portfolios of individual assets, we can further break down the tracking portfolio for the asset.

Interpretation (cont'd)

- We can create tracking portfolios from factor portfolios or from individual securities to track any investment.

	Portfolio Weights			
Tracking Portfolio	Security A	Security B	Security C	Sum
(2) Pure Factor 1	0.40	0.40	0.20	1
(1) Pure Factor 2	0.20	0.70	0.10	1
(-2) Risk-Free Port.	0.60	0.60	-0.20	1
(1) Asset $(= 2F_1 + F_2)$	-0.20	0.30	0.90	1

  - The position in the risk-free asset is 1- $\beta_1 - \beta_2$

load $\beta_1 F_1 + \beta_2 F_2$

$\beta_1 F_1 + \beta_2 F_2 + (1-\beta_1-\beta_2) F_3$

Interpretation (cont'd)

- If the tracking portfolio's expected return differs from the investment's, then there is **arbitrage**.

- We can buy the one with the higher expected return, and sell the one with the lower, and make money with no risk and no initial cash expenditure!

Arbitrage and Arbitrage Pricing

- Previously, we tracked the asset by forming a portfolio with the same factor beta configuration. This implied,

$$\tilde{R}_{asset} = \alpha_{asset} + 2\tilde{F}_1 + 1\tilde{F}_2$$
$$\tilde{R}_P = \alpha_P + 2\tilde{F}_1 + 1\tilde{F}_2$$

  - where portfolio P is the tracking portfolio
- Since both portfolios are **well-diversified**, idiosyncratic risk is eliminated.
- Thus, the only difference in the equations are the α's or expected returns.

Arbitrage and Arbitrage Pricing

- What happens if $\alpha_{asset} \neq \alpha_P$?

- Thus the expected return of the tracking portfolio is

$$\begin{aligned}
\alpha_P &= (1-\beta_1-\beta_2)E[R_{f*}] + \beta_1 E[R_{FP1}] + \beta_2 E[R_{FP2}] \\
\alpha_P &= (1-\beta_1-\beta_2)r_f + \beta_1(r_f+\lambda_1) + \beta_2(r_f+\lambda_2) \\
&= r_f + \beta_1\lambda_1 + \beta_2\lambda_2
\end{aligned}$$

- In our example, it is

$$\alpha_P = 0.02 + 2(0.03) + 1(0.02) = 0.10$$

Arbitrage and Arbitrage Pricing

- Suppose the expected return on the asset is 11%. Is there arbitrage? And, how do we exploit it?

- No matter what happens in the macroeconomy, we are perfectly hedged, since our arbitrage strategy takes exactly opposite positions on the factors, which are the sole source of risk for every investment.

Arbitrage Pricing Theory

- The use of tracking portfolios and the possibility to exploit arbitrage is the basis for the Arbitrage Pricing Theory (APT)

- The APT only requires three assumptions:
  1. Returns follow a factor model
  2. There are no arbitrage opportunities
  3. **There are many securities, so that we can form well-diversified portfolios that eliminate firm-specific risk**

Arbitrage Pricing Theory

- If we assume no arbitrage, and returns follow a factor model, then every asset must have the same expected return as its tracking portfolio.

- The expected return of the tracking portfolio, α_P, is

$$r_f + \beta_{i1}\lambda_1 + \ldots + \beta_{iK}\lambda_K$$

  - where the λ's are the risk premiums on the K factors,

APT equation:

- Then, we have the APT equation:

$$E[\tilde{r}_i] = \alpha_i = \alpha_P = r_f + \beta_{i1}\lambda_1 + \ldots + \beta_{iK}\lambda_K$$

  - Reminder: In CAPM: $\mathrm{E}[r_i] = r_f + \beta_i\left(\mathrm{E}[r_m] - r_f\right)$

- Thus, we now have a statement about expected returns. In other words, we now have a pricing model.

CAPM vs. APT

- Note: unlike the CAPM, the only assumption here is that no arbitrage exists. APT requires less assumption.
- Thus, the APT is much less restrictive than the CAPM.
- However, this comes at a cost, because the APT is also **less specific**.
  - the APT does not tell us how many factors
  - or what those factors are 3 need to figure out (Difficult)
  - the CAPM tells us there is one factor, and it is the market portfolio

General Ideas in Asset Pricing

- A security's expected return depends on when it makes its payments
- A security is attractive if it pays up when times are bad
  - that's exactly when you need the money!
  - you are prepared to accept a *low* expected return on these securities
- A security is not attractive if it pays up when times are good
  - you don't need any more money at these times
  - you require a high expected return on these securities

General Ideas in Asset Pricing

- In the CAPM, good and bad times are captured by the market return
  - Good times: market return is high, and vice-versa
- Securities with a high beta tend to pay up (i.e., earn high returns) when the market is doing well
  - What use is that?
  - The security is unattractive
  - It must earn a high expected return

Summary

- The assumptions for APT
  - Asset returns have a factor structure
  - No arbitrage exists.
- APT does not tell us
  - how many factors
  - what those factors are